

Individuals at Risk for Vitamin A Toxicity

Individuals most at risk for vitamin A toxicity include patients taking systemic vitamin A derivatives for the treatment of dermatologic conditions such as **acne**, **psoriasis**, and **ichthyosis**. Other at-risk populations include vitamin food faddists who consume large quantities of nonprescription vitamin A supplements. Two notable episodes of vitamin A toxicity occurred in the 1950s, when very high levels of vitamin A were added to infant formulas, and in the 1970s, when high doses of vitamin A were used to treat various dermatologic diseases.

Clinical Findings

Acute Vitamin A Toxicity

Individuals with acute vitamin A toxicity present with: - Dry, scaly skin - Widespread desquamation - Fissuring of the lips and angles of the mouth - Headache - Fatigue - Anorexia - Nausea and vomiting - Blurred vision - Pseudotumor cerebri - Myalgias and arthralgias

Chronic Vitamin A Toxicity in Adults

Early cutaneous signs include dryness of the lips, which may progress to: - Diffuse, dry, pruritic, scaly skin - Peeling of palms and soles - Alopecia - Follicular hyperkeratosis - Hyperpigmentation of the face and neck

Systemic symptoms may include anorexia, fatigue, and weight loss. Notably, **follicular hyperkeratosis** can occur in both vitamin A deficiency (VAD) and toxicity.

Chronic Vitamin A Toxicity in Children

Children may present with: - Coarse hair and diffuse alopecia - Coarse skin with generalized exfoliation - Hyperpigmentation - Exfoliative cheilitis - Bulging fontanelle (due to pseudotumor cerebri) - Headaches and papilledema

Skeletal manifestations are common and include: - Growth retardation due to premature closure of epiphyses - Spontaneous bone fractures

Proposed mechanisms for bone pathology involve antagonism between vitamin A- and vitamin D-mediated intracellular signaling pathways and interactions with calcium-regulating hormones.

Long-Term Effects

Deposition of excess vitamin A in adipose tissue and perisinusoidal fibrosis of the liver can lead to **cirrhosis**, representing the most significant long-term consequences of chronic toxicity.

Laboratory Testing

Findings in hypervitaminosis A include: - Elevated serum vitamin A levels - Increased serum calcium - Elevated alkaline phosphatase (AP)

Altered calcium homeostasis may result in calcification of tendons, ligaments, and soft tissues. Liver involvement, including perisinusoidal fibrosis progressing to cirrhosis, is a major concern in prolonged cases.

Treatment

Most symptoms of vitamin A toxicity resolve after discontinuation of excess vitamin intake. However, **liver cirrhosis** and complications from **pseudotumor cerebri** may persist.

Carotenemia and Carotenoderma

Epidemiology

Although hypervitaminosis A causes a wide range of clinical effects, excessive carotene intake leads to a benign condition characterized by yellow-orange skin pigmentation. This condition, termed **carotenemia**, was first described in 1919 by Hess and Meyers, who linked yellow skin discoloration with elevated serum carotene levels. Carotenemia became more prevalent during World War I and World War II due to dietary shifts from meat-based to plant-based diets caused by food shortages.

Etiology and Pathogenesis

Carotenes are not synthesized in the body and must be obtained through diet. Plant carotenes are converted to vitamin A in the gastrointestinal tract, but about one-third is absorbed directly.

Factors influencing carotene absorption include: - **Thyroid hormone levels:** Hypothyroidism reduces conversion of carotene to retinol, leading to elevated serum carotene - **Pancreatic lipase and bile acids:** Deficiencies due to pancreatic, biliary, or hepatic disease impair carotene digestion and increase levels - **Food processing:** Mashing, cooking, and pureeing increase carotene availability by disrupting plant cell membranes - **Dietary fat:** Enhances absorption (carotene is fat-soluble) - **Dietary fiber:** Decreases carotene absorption